Why Does a Small CNN Underperform Random Forests? A Small-Data Diagnostic Study of Kepler DR25 Transit Vetting

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Abstract

Deep networks are the standard approach to vetting Kepler Threshold Crossing Events (TCEs), yet on a few thousand labeled examples a small 1D CNN falls well short of a Random Forest. We study why, using 4,515 preprocessed DR25 TCEs from 3,724 stars, star-grouped splits that prevent leakage between training and test sets, and five seeds for every configuration. A Random Forest on four catalog statistics plus nine flux-shape features reaches F1 = 0.884 ± 0.004 (ROC-AUC 0.942), against 0.806 ± 0.018 (0.878) for a 3,953-parameter CNN on the phase-folded flux and 0.713 for logistic regression on the catalog statistics; grouped five-fold cross-validation agrees (0.901 vs. 0.821). Feature-set ablations locate the gap in the inputs: a Random Forest on the catalog statistics alone reaches 0.880 ± 0.003, one on the flux features alone reaches 0.765 ± 0.002, below the CNN, and giving the CNN the same four catalog statistics closes the gap (F1 0.889 ± 0.004, ROC-AUC 0.939). Training-set size, noise augmentation, a learning-rate-schedule control and a dual-view branch do not close it. At the natural class ratio of roughly 1:7, F1 falls to 0.649 for the Random Forest and 0.611 for the CNN with catalog statistics.

1 Introduction

The Kepler space telescope monitored the brightness of over 150,000 stars for four years, searching for the periodic dimming caused by a planet transiting its host star (Borucki et al., 2010). Kepler's automated pipeline flags any statistically significant periodic dimming as a Threshold Crossing Event (TCE); the final Data Release 25 (DR25) catalog contains 34,032 such TCEs (Thompson et al., 2018). The great majority are not planets: they are caused by eclipsing binary stars, instrumental artifacts, or background contamination. NASA's Robovetter, a rule-based decision system, dispositions each TCE as a Planet Candidate (PC) or False Positive (FP); of the TCEs with a Robovetter disposition, fewer than one in eight are planet candidates.

A real transit produces a shallow, symmetric, consistently timed dip, while many false positives produce irregular or asymmetric ones. This has made TCE vetting an active application area for machine learning, from random-forest classifiers on pipeline diagnostics (McCauliff et al., 2015) to the AstroNet convolutional architecture of Shallue and Vanderburg (2018). Deep vetting systems are typically trained on larger sets of labeled TCEs with dual local/global input views and, in later work, additional domain information.

We ask a narrower question on a modestly sized dataset (4,515 successfully preprocessed lightcurves): when a compact 1D CNN trained on raw folded flux falls behind classical models on hand-engineered features, *why*? We first establish that the gap is real under leakage-free, multi-seed evaluation. We then test candidate explanations one at a time: training-set size (learning curves), the training recipe (noise augmentation and a learning-rate schedule), the information each model receives (feature-set ablations, including a CNN given the catalog statistics), the input representation (a dual-view variant and the number of in-transit samples in the folded array), and the type of false positive each model misclassifies.

2 Related Work

McCauliff et al. (2015) trained a random-forest classifier on diagnostic attributes of Kepler TCEs to sort them into planet candidates, astrophysical false positives and systematic noise.Shallue and Vanderburg (2018) introduced

AstroNet, a convolutional network trained on global and local views of phase-folded Kepler lightcurves, which remains the reference deep-learning architecture for this task. Ansdell et al. (2018) showed that adding domain knowledge to this architecture, namely centroid time series and stellar parameters, improved accuracy from 95.8% to 97.5%. Valizadegan et al. (2022) proposed ExoMiner, a deep classifier built around the diagnostic tests that human vetters apply, and used it to validate 301 new planets; Armstrong et al. (2021) used a Gaussian-process classifier and other models to validate 50 Kepler planets. Closest to our feature-engineering comparison, Malik et al. (2022) extracted 789 time-series features per lightcurve with TSFresh and trained a LightGBM classifier, obtaining performance on Kepler data comparable to deep models.

Our engineered flux-shape features are a much smaller, from-scratch analogue of such feature sets, computed from our own 200-point folded arrays. Our contribution is not a new classifier but a controlled diagnosis of why a small CNN on this representation underperforms a feature-based model at this data scale.

3 Data and Preprocessing

3.1 Catalog construction

We queried the NASA Exoplanet Archive TAP API for all 34,032 TCEs in the `q1_q17_dr25_tce` table, retrieving four catalog-derived statistics per TCE: orbital period, transit duration, transit depth, and model signal-to-noise ratio (SNR). We merged this catalog with NASA's DR25 Robovetter output (disposition and confidence score) on (`kepid`, `tce_plnt_num`), since a single star can host multiple TCEs and both identifiers are needed to uniquely match an event. This yielded 32,534 TCEs with both features and labels (28,497 FP, 4,037 PC), matching the overall scale reported by Thompson et al. (2018) for the DR25 catalog.

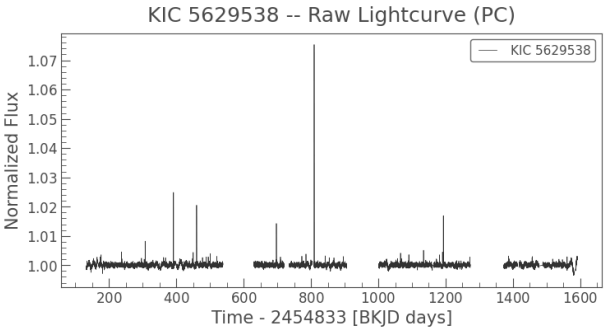
To manage the roughly 7:1 class imbalance during model development, we constructed a balanced subset by taking all 4,037 planet candidates and randomly sampling 4,037 false positives (seed 42), for 8,074 TCEs total.

3.2 Lightcurve retrieval and flux preprocessing

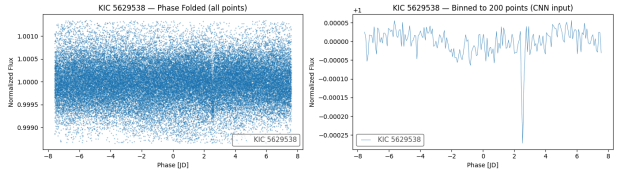
For each of the 8,074 balanced TCEs, we used the `lightkurve` package to download all available long-cadence lightcurve quarters from NASA MAST, stitch them into a single time series, and apply the following preprocessing:

1. **Sigma-clipping** at 3 standard deviations to remove outlier flux spikes.
2. **Phase-folding** on the TCE's catalog period, anchored to its catalog epoch (`tce_time0bk`) so the transit consistently lands at the center of the folded array rather than at an arbitrary phase.
3. **Binning** to a fixed length of 200 points.
4. **Z-score normalization** (zero mean, unit variance) per star.

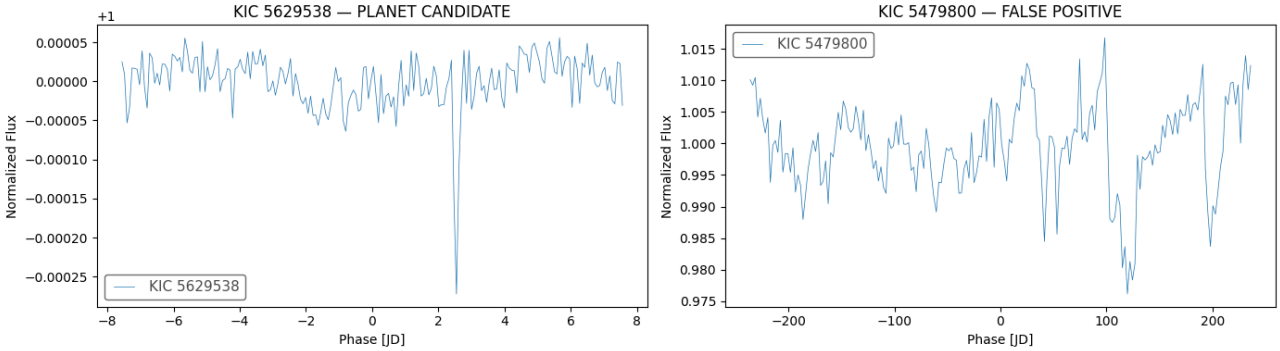
This pipeline succeeded for 4,515 of the 8,074 targeted TCEs (55.9%); failures, primarily missing or malformed MAST downloads, were logged and excluded rather than imputed. The resulting set is close to balanced (2,328 PC, 2,187 FP; ratio 0.94:1). Figure 1 shows a representative planet candidate and false positive at each stage of this pipeline.



(a) Raw stitched lightcurve, KIC 5629538 (PC).



(b) Same star: phase-folded, then binned to 200 points.



(c) A planet candidate (left) shows a clean, symmetric transit dip; a false positive (right, KIC 5479800) does not.

Figure 1: Preprocessing pipeline and PC/FP visual comparison.

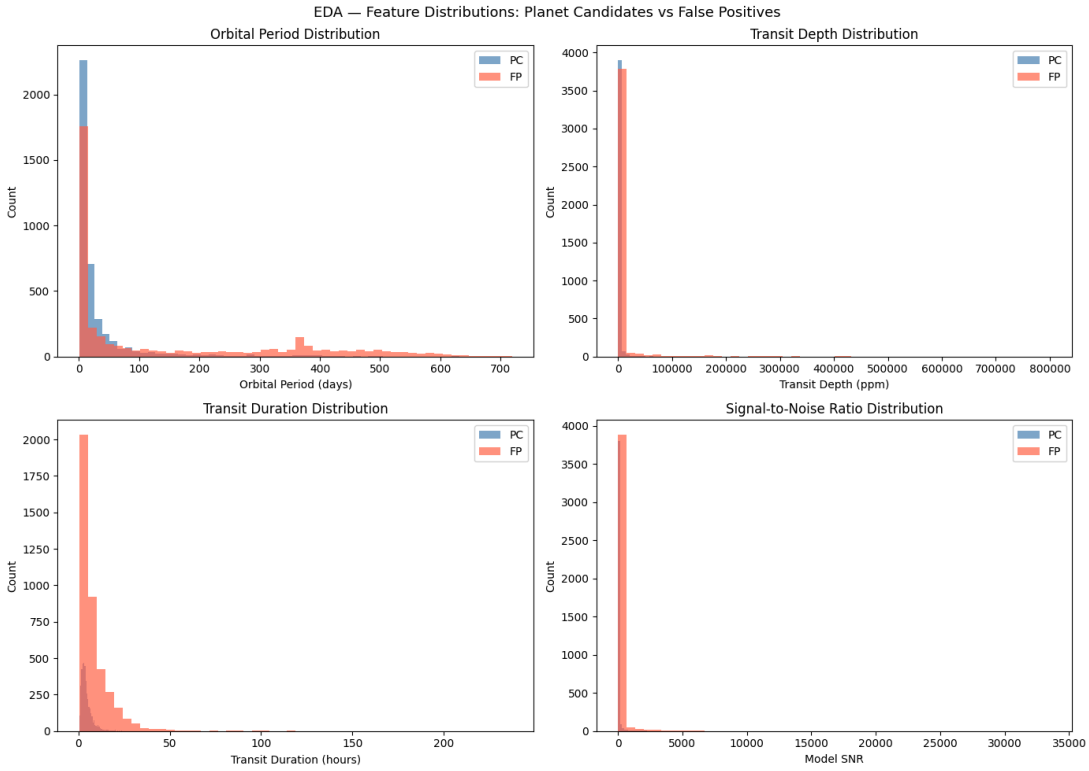


Figure 2: Distributions of the four catalog-derived statistics (orbital period, transit depth, transit duration, model SNR) for planet candidates versus false positives in the balanced 8,074-TCE set. All four are heavily right-skewed with substantial PC/FP overlap, motivating the use of a nonlinear classifier over these features.

3.3 Transit coverage of the folded array

Because each array spans the full orbital phase in 200 bins, the number of bins that fall in transit is 200 × (transit duration / orbital period). Across the 4,515 TCEs this has a median of 2.3 bins (interquartile range 0.6–6.7), and 69.5% of TCEs have five or fewer bins in transit. For most TCEs the transit is therefore described by only a few samples, which bounds what any model reading the folded array alone can learn about transit shape.

3.4 Star-grouped train/validation/test split

The 4,515 TCEs come from 3,724 stars; 592 stars host more than one TCE (1,383 TCEs, 30.6%). TCEs of the same star share stellar properties and instrumental systematics, so a row-level split lets a model see a star in training and be tested on a sibling TCE. In an ungrouped stratified 70/15/15 split of this data, 143 to 183 of the test TCEs (mean 166, over 20 row orders) share a star with the training set.

We therefore split with `StratifiedGroupKFold` (groups = `kepid`, seed 42): the TCEs are partitioned into 20 star-disjoint, label-stratified folds, of which 3 form the test set, 3 the validation set and 14 the training set. Groups are indivisible, so partition sizes are approximate: 3,148 training TCEs (1,627 PC / 1,521 FP), 706 validation, and 661 test (329 PC / 332 FP). No star appears in more than one partition (asserted in code). All models are trained and evaluated on this split unless otherwise noted (Section 4.1).

4 Methods

We compare four models and three feature-set ablations, all evaluated with the same metrics (accuracy, precision, recall, F1, ROC-AUC) on the same test set. Neural models and the Random Forests are run with five seeds (42–46); the logistic regression is deterministic and run once. We report mean ± standard deviation across seeds.

LR_catalog. A logistic regression classifier trained on the four catalog-derived statistics alone (orbital period, transit duration, transit depth, model SNR), standardized before fitting. No transit search is run: the four statistics are the summary outputs of the Kepler pipeline's transit search, and the model is a linear classifier over them.

RF_main. A Random Forest (300 trees) trained on the four catalog statistics plus nine features we engineer directly from each star's 200-point folded flux array: mean, standard deviation, minimum, maximum, an estimated transit depth (out-of-transit mean minus in-transit minimum, using the central third of the array as a crude in-transit window), in-transit standard deviation, out-of-transit standard deviation, skewness, and kurtosis. This gives RF_main 13 features, none of which use Robovetter output. The seed sets the forest's randomness.

RF_with_robovetter. Identical to RF_main but with the Robovetter's confidence score added as a 14th feature. Since that score comes from the system that assigned each TCE's ground-truth label, we report this model only as an upper-bound reference, not as a baseline.

RF_catalog_only and RF_flux_only. Ablations of RF_main that isolate where its performance comes from: the same Random Forest trained on only the four catalog statistics (RF_catalog_only) or only the nine engineered flux features (RF_flux_only).

CNN_1D. A 1D convolutional network trained directly on the 200-point normalized flux array, with no catalog or engineered features. The architecture is three convolutional blocks (8, 16, and 32 filters; kernel size 5; padding 2) each followed by batch normalization, ReLU, max pooling (factor 2), and dropout (0.3), then global average pooling and a two-layer classification head (32→16→1). We use global average pooling rather than flattening to limit parameter count relative to dataset size: an earlier version with 16/32/64 filters and a flattened head had 15,329 trainable parameters, which we judged likely to overfit a training set of about 3,000 examples; the reported architecture has 3,953. The model is trained with `BCEWithLogitsLoss` and AdamW (learning rate 10^{-3}, weight decay 10^{-4}, batch size 64), with early stopping on validation loss (patience 8, maximum 60 epochs) as a third regularizer alongside dropout and weight decay. The seed sets weight initialization and mini-batch order. Figure 4 shows the training curves of the seed-42 run.

CNN_1D_catalog. The ablation that gives the CNN the same inputs as RF_catalog_only: CNN_1D with the four catalog statistics concatenated to the 32-dimensional pooled convolutional features before the classification head (4,017 parameters). The statistics are passed through a signed log transform (model SNR contains -1 sentinel values) and standardized with the training-set mean and standard deviation. Training protocol and seeds are those of CNN_1D.

4.1 Cross-validation protocol

To test whether the RF_main/CNN_1D gap depends on one particular split, we also ran 5-fold `StratifiedGroupKFold` cross-validation (groups = `kepid`, seed 42) over all four models on the full 4,515-TCE set. Within each fold, the CNN uses a further 85/15 stratified split of that fold's training TCEs for early-stopping validation, with the same training budget as the single-split run, so no test-fold data is used during training. Each fold uses one seed (42).

4.2 Natural-prevalence evaluation

Our test set is balanced, but in the full catalog planet candidates are about 12.4% of TCEs (4,037 of 32,534). Recall and false-positive rate do not depend on class prevalence, so we re-weight to the natural prevalence $\pi = 0.124$ analytically from each run's measured recall R and false-positive rate FPR: $\text{precision}_{nat} = \pi R / (\pi R + (1-\pi)\text{FPR})$, and F1 and accuracy follow from it. ROC-AUC is unchanged.

4.3 Augmentation sweep

We test whether per-sample additive Gaussian noise, applied only to the CNN's training data, improves generalization, sweeping the noise strength α in 0, 0.05, 0.1, 0.2, 0.3, 0.5 with five seeds per level. Noise is scaled per sample to the sample's own flux standard deviation ($\text{noise}_std = \alpha \cdot \sigma_{sample}$), since transit depths vary substantially across TCEs. Noise is redrawn independently for every mini-batch and never applied to validation or test data. The α=0 condition is the CNN_1D runs themselves.

4.4 Error-mode analysis

To characterize *what kind* of false positive each model struggles with, we use two diagnostics present in the DR25 catalog but not used as model inputs: an independent autovetter's vote fractions for “not transit-like” (`av_vf_ntp`) and “astrophysical false positive” (`av_vf_afp`), and the odd/even transit-depth comparison statistic (`tce_bin_oedp_stat`), whose magnitude is the standard signature of an eclipsing binary masquerading as a planet. For each false positive in the test set, we assign a heuristic category by taking whichever of these three signals is most extreme relative to its own top-tertile threshold among test-set false positives, then compare each model's error rate within each category. Because `tce_bin_oedp_stat` is unbounded while the two vote fractions lie in [0,1], raw magnitudes are not comparable across the three signals; in practice the odd-even signal dominates the categorization whenever it clears its threshold, leaving the “not transit-like” and “astrophysical false positive” categories empty. We therefore report only odd-even-flagged (likely eclipsing binary) versus unflagged false positives. Error rates are computed per seed and averaged.

4.5 Dual-view CNN and ensemble

Following the local/global input design of Shallue and Vanderburg (2018), we add a second branch that sees only a 40-point window cropped from the center of the same 200-point array, where phase-folding places the transit. This is a weaker version of the original design: a true local view is re-derived from raw lightcurves at finer time resolution around the transit, whereas cropping an already coarse array adds no resolution. Each branch is a two-block convolutional stack (8 then 16 filters, kernel 5) with global average pooling; the two 16-dimensional branch outputs are concatenated and passed through the same two-layer head as CNN_1D, giving 2,049 trainable parameters. The dual-view model is trained with a cosine-annealed learning rate. Because this changes the schedule as well as the architecture, we also train CNN_1D with the same cosine schedule as a control. We train

25 dual-view models (seeds 42–66) and form five ensembles from disjoint groups of five by averaging predicted probabilities.

4.6 Learning curves

To test whether the gap reflects a data-quantity limitation, we train RF_main and CNN_1D on stratified subsamples of the training set (10%, 25%, 50%, 75%, 100%), five seeds each (the seed also selects the subsample), and evaluate on the same fixed test set throughout. The validation set is unchanged.

5 Results

5.1 Single-split comparison

Table 1 reports all four models on the shared 661-TCE test set. RF_main is the strongest model without access to expert labels (F1 = 0.884 ± 0.004), followed by CNN_1D (0.806 ± 0.018) and LR_catalog (0.713). RF_with_robovetter is an upper-bound reference only (Section 4). The CNN's seed-to-seed standard deviation in F1 (0.018) is more than four times the Random Forest's (0.004).

Table 1: Test-set results (n=661), mean ± std over five seeds; LR_catalog is deterministic.

Model	Features	Accuracy	Precision	Recall	F1	ROC-AUC
LR_catalog	4	0.673	0.634	0.815	0.713	0.717
CNN_1D	200 (raw flux)	0.810 ± 0.023	0.824 ± 0.037	0.789 ± 0.010	0.806 ± 0.018	0.878 ± 0.012
RF_main	13	0.884 ± 0.003	0.880 ± 0.004	0.888 ± 0.007	0.884 ± 0.004	0.942 ± 0.001
RF_with_robovetter*	14	0.991 ± 0.000	0.994 ± 0.001	0.987 ± 0.001	0.991 ± 0.000	0.999 ± 0.000

*Includes the Robovetter's own confidence score as a feature; upper-bound reference, not a baseline.

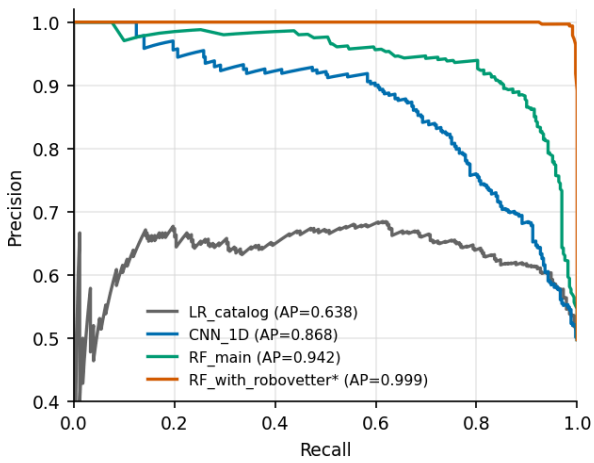
Because the test set is balanced while real TCE populations are not, Table 2 re-weights precision, F1 and accuracy to the natural prevalence (Section 4.2). Absolute numbers drop sharply for every model, because a given false-positive rate now acts on about seven times more false positives. RF_main's precision falls to 0.512 ± 0.009 (F1 0.649 ± 0.007) and CNN_1D's to 0.405 ± 0.056 (F1 0.533 ± 0.051); the ranking of models is unchanged.

Table 2: Metrics re-weighted to the natural PC prevalence (12.4%), mean ± std over five seeds.

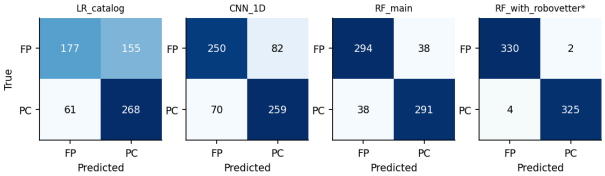
Model	F1 (balanced)	Precision (nat.)	F1 (nat.)	Accuracy (nat.)
LR_catalog	0.713	0.198	0.319	0.568
CNN_1D	0.806 ± 0.018	0.405 ± 0.056	0.533 ± 0.051	0.826 ± 0.040
RF_main	0.884 ± 0.004	0.512 ± 0.009	0.649 ± 0.007	0.881 ± 0.004
RF_with_robovetter*	0.991 ± 0.000	0.963 ± 0.009	0.975 ± 0.004	0.994 ± 0.001
CNN_dualview	0.799 ± 0.005	0.439 ± 0.020	0.555 ± 0.015	0.849 ± 0.010
CNN_1D + catalog	0.889 ± 0.004	0.456 ± 0.031	0.611 ± 0.026	0.852 ± 0.019

*Includes the Robovetter's own confidence score as a feature; upper-bound reference, not a baseline.

Figure 3 shows precision-recall curves and confusion matrices on this test set for the seed-42 run of each model.



(a) Precision-recall curves.



(b) Confusion matrices.

Figure 3: Seed-42 runs on the test set. Left: precision-recall curves with average precision (AP) in the legend. Right: confusion matrices (FP=0, PC=1). LR_catalog catches 268 of 329 PCs but misclassifies 155 of 332 FPs as PC; CNN_1D catches 259 PCs and misclassifies 82 FPs; RF_main catches 291 PCs and misclassifies 38 FPs.

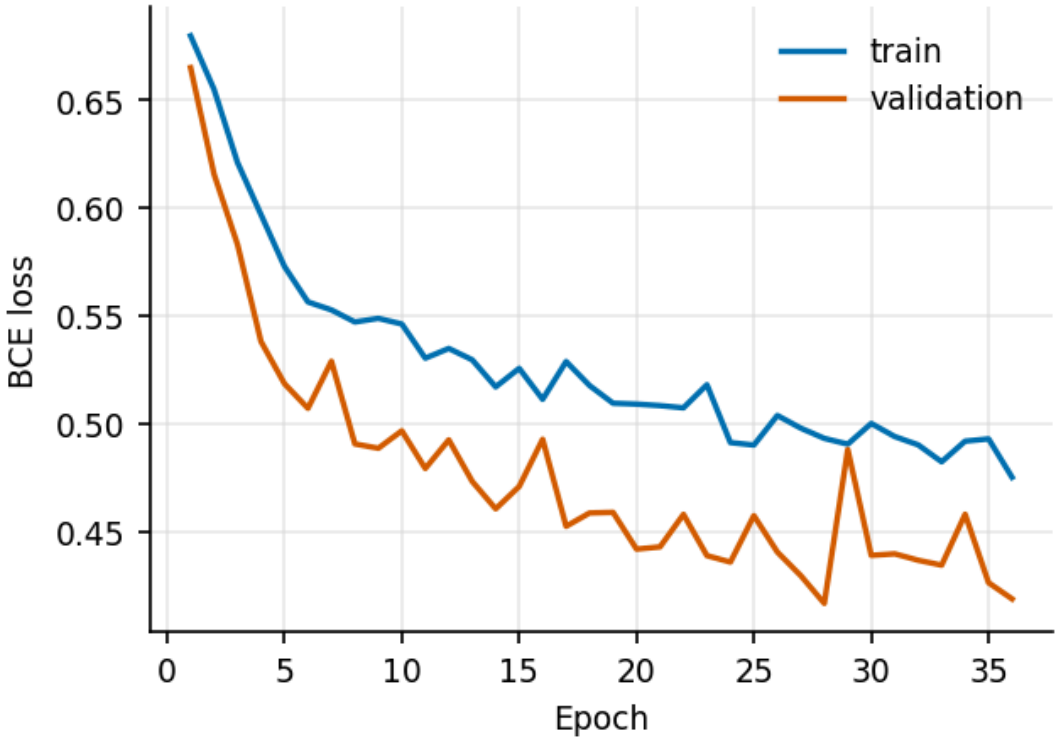


Figure 4: CNN_1D training curves (seed-42 run). Training and validation loss track closely with no divergence, consistent with dropout, weight decay and early stopping controlling overfitting at this dataset size.

5.2 Cross-validated comparison

Table 3 reports grouped 5-fold cross-validated mean ± standard deviation. The RF_main/CNN_1D gap persists: mean F1 is 0.901 vs. 0.821, a difference of 0.080, more than four times either model's fold-to-fold standard deviation (±0.018 and ±0.013). The five-seed single-split CNN_1D mean (0.806) lies within one seed standard deviation of its cross-validated mean (0.821), so the single split is not unusually unfavorable to the CNN.

Table 3: Grouped 5-fold cross-validation results (mean ± std across folds).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR_catalog	0.695 ± 0.009	0.652 ± 0.013	0.874 ± 0.021	0.747 ± 0.014	0.722 ± 0.006
CNN_1D	0.818 ± 0.008	0.833 ± 0.028	0.812 ± 0.039	0.821 ± 0.013	0.891 ± 0.005
RF_main	0.899 ± 0.017	0.906 ± 0.009	0.897 ± 0.033	0.901 ± 0.018	0.956 ± 0.007
RF_with_robovetter*	0.994 ± 0.004	0.994 ± 0.005	0.994 ± 0.006	0.994 ± 0.004	0.999 ± 0.001

*Includes the Robovetter's own confidence score as a feature; upper-bound reference, not a baseline.

5.3 Feature importance

Figure 5 shows RF_main's feature importances (mean over five seeds). The most important feature is the catalog's model SNR (0.205 ± 0.006), consistent with the intuition that a higher-confidence transit detection is more likely to be a real planet. The second is `flux_depth_est` (0.148 ± 0.005), one of the nine features we engineered from the folded flux array, and three of the five most important features (`flux_depth_est`, `flux_min`, `flux_skew`) are engineered. Importance does not imply marginal predictive value, since correlated features share importance; Section 5.4 measures the contribution directly.

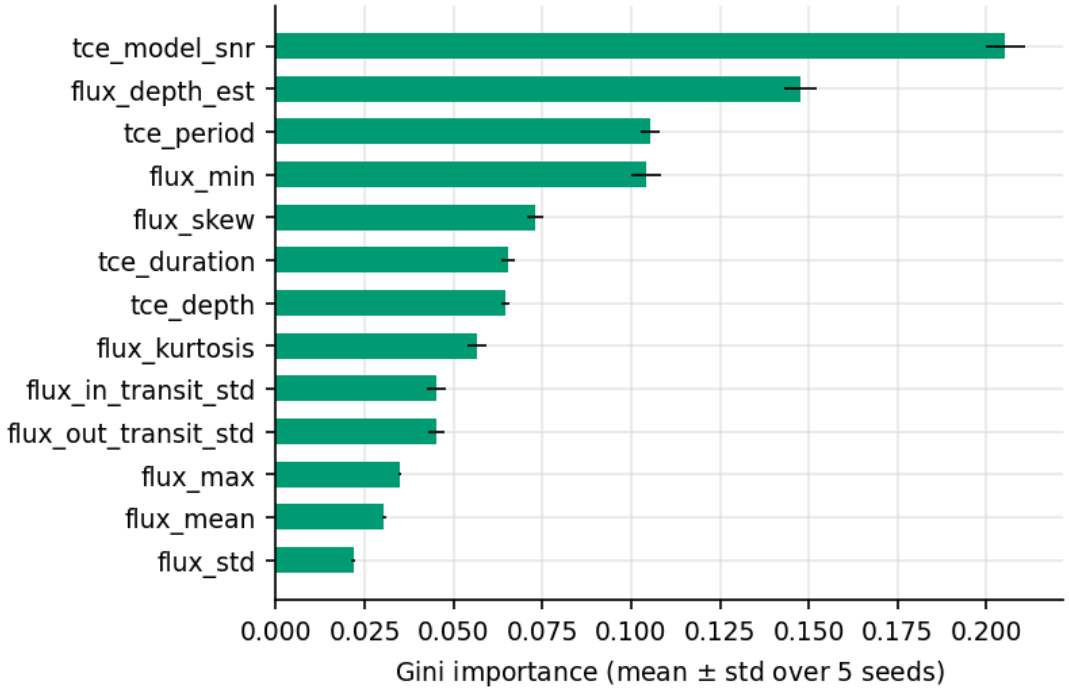


Figure 5: RF_main feature importances (Gini importance, mean ± std over five seeds). Catalog features (`tce_*`) and engineered flux-shape features (`flux_*`) are interleaved in importance.

5.4 Feature-set ablation

Table 4 separates the contribution of the catalog statistics from that of the flux. A Random Forest on the four catalog statistics alone reaches F1 = 0.880 ± 0.003 and ROC-AUC 0.935, close to RF_main (0.884 ± 0.004; 0.942): the nine engineered flux features add under 0.004 F1 and about 0.007 ROC-AUC. A Random Forest on the engineered flux features alone reaches only 0.765 ± 0.002 (ROC-AUC 0.853), below CNN_1D (0.806 ± 0.018; 0.878), so given the folded flux the CNN extracts more than our hand-engineered statistics do. The decisive control is to give the CNN the catalog statistics: CNN_1D_catalog reaches F1 = 0.889 ± 0.004 and ROC-AUC 0.939 ± 0.002, matching RF_main. Its F1 is higher than RF_main's in four of five same-seed pairs (by 0.004 to 0.011) and its ROC-AUC is lower in all five (by 0.001 to 0.008), and its seed-to-seed F1 standard deviation falls from 0.018 to 0.004. Relative to the catalog-only Random Forest, adding the flux gives the CNN a small gain (F1

0.889 vs. 0.880; ROC-AUC 0.939 vs. 0.935). The gap between the CNN and the Random Forest is therefore a gap in available inputs, not in the learner. The nonlinear model also matters: logistic regression on the same four statistics reaches only 0.713.

The match is at the default 0.5 threshold on a balanced test set. At the natural class ratio, CNN_1D_catalog's F1 is 0.611 ± 0.026 against 0.649 ± 0.007 for RF_main (Table 2), because it trades precision (0.854 vs. 0.880 on the balanced set) for recall (0.928 vs. 0.888), and extra false alarms weigh more when false positives are seven times as common.

Table 4: Feature-set ablation, test set, mean ± std over five seeds; LR_catalog is deterministic.

Model	Inputs	Features	F1	ROC-AUC
LR_catalog	catalog statistics	4	0.713	0.717
RF_catalog_only	catalog statistics	4	0.880 ± 0.003	0.935 ± 0.000
RF_flux_only	engineered flux features	9	0.765 ± 0.002	0.853 ± 0.001
RF_main	catalog + engineered flux	13	0.884 ± 0.004	0.942 ± 0.001
CNN_1D	folded flux array	200	0.806 ± 0.018	0.878 ± 0.012
CNN_1D_catalog	folded flux + catalog statistics	204	0.889 ± 0.004	0.939 ± 0.002

5.5 Calibration and significance

Figure 6 shows reliability diagrams for the seed-42 runs and Table 5 reports Brier scores (lower is better). RF_main has a Brier score of 0.091 ± 0.001 and CNN_1D 0.138 ± 0.010. In the seed-42 reliability diagram, CNN_1D sits slightly below the diagonal for mean predicted probabilities between about 0.4 and 0.75, but neither model is strongly miscalibrated; the Brier difference mostly reflects the models' differing discrimination.

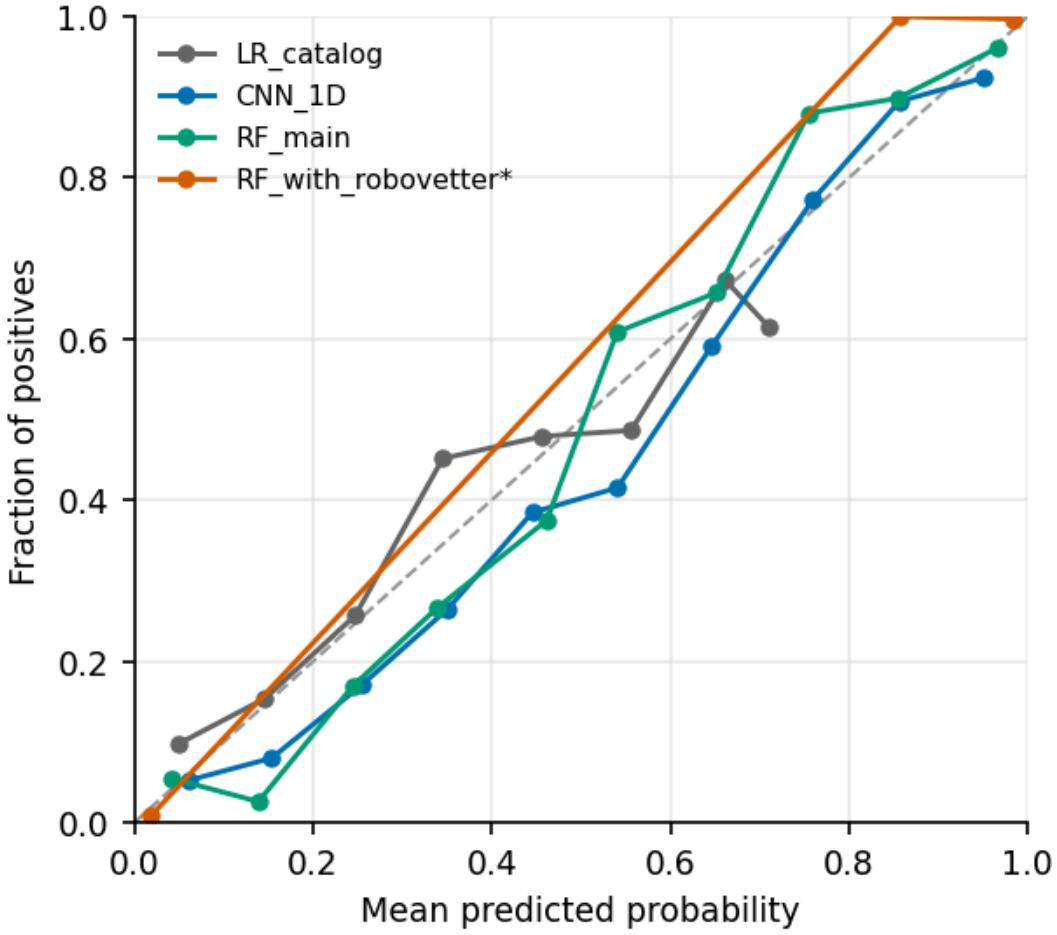


Figure 6: Reliability diagram for the seed-42 runs, test set (n=661), ten equal-width probability bins (bins with fewer than five examples omitted).

Table 5: Brier score (lower is better), test set, mean ± std over five seeds.

Model	Brier score
LR_catalog	0.207
CNN_1D	0.138 ± 0.010
RF_main	0.091 ± 0.001
RF_with_robovetter*	0.008 ± 0.000

*Includes the Robovetter's own confidence score as a feature; upper-bound reference, not a baseline.

We also test whether the RF_main/CNN_1D gap is distinguishable from chance. McNemar's exact test on paired test-set predictions of the two models trained with the same seed gives $p < 5 \times 10^{-5}$ for every one of the five seed pairs: RF_main correctly classifies 63 to 100 TCEs that CNN_1D misses, while CNN_1D correctly classifies 22 to 30 that RF_main misses.

5.6 Augmentation sweep

Figure 7 and Table 6 report CNN_1D performance across six noise levels with five seeds each. Noise augmentation gives no reliable F1 benefit: at $\alpha = 0.05$ and 0.1 mean F1 is 0.814 and 0.810 against 0.806 without noise, differences smaller than the seed standard deviation (0.008 to 0.018). At larger noise F1 falls (0.797, 0.793 and 0.786 at $\alpha = 0.2$, 0.3, 0.5), driven mainly by recall, which declines steadily from 0.789 to 0.740 while precision stays roughly flat. Only $\alpha = 0.5$ has a clearly lower ROC-AUC (0.853 ± 0.016 vs. 0.878 ± 0.012), and its F1 has the widest seed spread (0.028). Noise scaled to each sample's own flux variance can be large relative to a

shallow transit, so one plausible reading is that strong noise partly obscures the shape signal; we did not test this directly, nor noise levels below α = 0.05.

Table 6: CNN_1D performance across augmentation noise levels α (test set, mean ± std over five seeds).

α	Accuracy	Precision	Recall	F1	ROC-AUC
0.00 (no aug.)	0.810 ± 0.023	0.824 ± 0.037	0.789 ± 0.010	0.806 ± 0.018	0.878 ± 0.012
0.05	0.821 ± 0.014	0.845 ± 0.029	0.786 ± 0.020	0.814 ± 0.013	0.884 ± 0.009
0.10	0.819 ± 0.010	0.848 ± 0.028	0.776 ± 0.019	0.810 ± 0.008	0.882 ± 0.008
0.20	0.808 ± 0.015	0.840 ± 0.033	0.759 ± 0.018	0.797 ± 0.012	0.876 ± 0.010
0.30	0.808 ± 0.012	0.853 ± 0.023	0.742 ± 0.020	0.793 ± 0.012	0.869 ± 0.012
0.50	0.798 ± 0.032	0.839 ± 0.053	0.740 ± 0.013	0.786 ± 0.028	0.853 ± 0.016

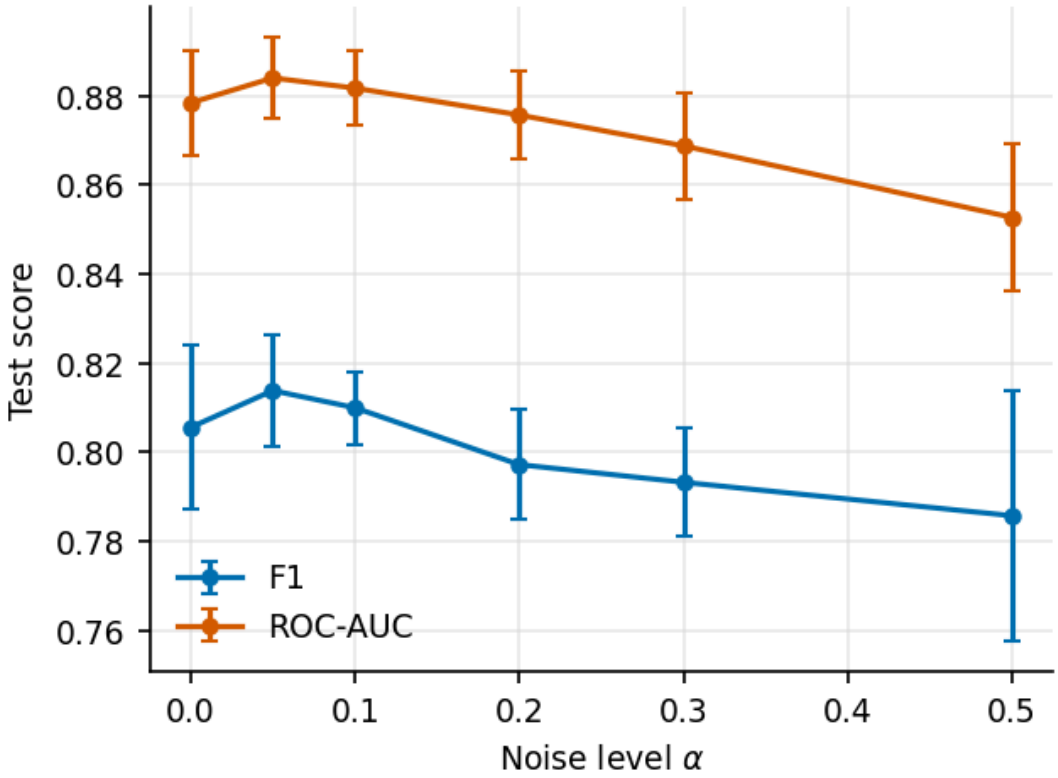


Figure 7: CNN_1D F1 and ROC-AUC across six augmentation noise levels (mean ± std over five seeds).

5.7 Error-mode analysis

Table 7 and Figure 8 report each model's error rate on test-set false positives by heuristic category (Section 4.4; only the odd-even-flagged and unflagged categories are non-empty). On the 111 likely-eclipsing-binary false positives, CNN_1D's error rate (0.182 ± 0.046) is about 1.9 times RF_main's (0.097 ± 0.010). On the 221 unflagged or ambiguous false positives the gap is smaller (0.163 ± 0.045 vs. 0.131 ± 0.007) and lies within the CNN's seed spread. The CNN's excess errors are thus concentrated, though not exclusively, on likely eclipsing binaries. We do not test why: the folded array averages odd and even transits, so neither model sees odd and even depths separately, and the odd-even flag may correlate with other shape properties that RF_main's depth and minimum-flux features capture.

Table 7: Error rate on test-set false positives by heuristic category (see Section 4.4); mean ± std over five seeds.

Category	n	RF_main error rate	CNN_1D error rate
Odd-even flagged (likely eclipsing binary)	111	0.097 ± 0.010	0.182 ± 0.046
Unflagged / ambiguous	221	0.131 ± 0.007	0.163 ± 0.045

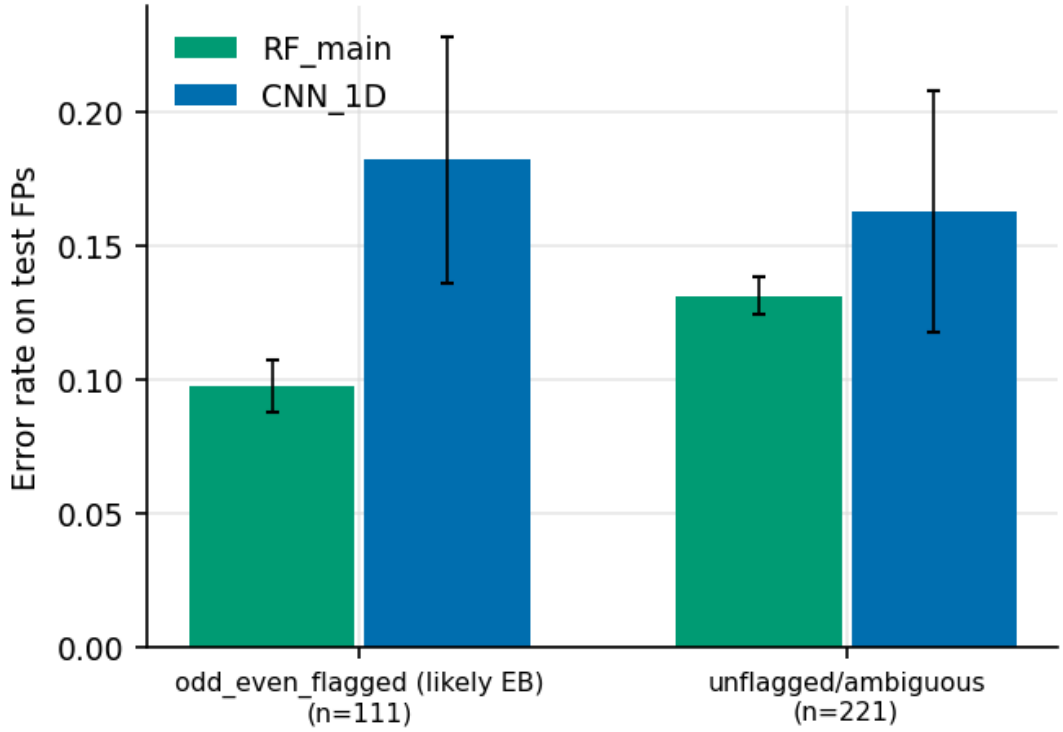


Figure 8: Error rate by heuristic false-positive category (mean ± std over five seeds).

5.8 Learning curves

Figure 9 and Table 8 show test F1 for RF_main and CNN_1D as a function of training-set size. RF_main rises from 0.842 ± 0.004 (10% of training data, n = 314) to 0.884 ± 0.004 (100%, n = 3,148), a gain of 0.042. CNN_1D rises from 0.753 ± 0.010 to 0.806 ± 0.018, a gain of 0.053, but its mean is flat over the last quarter of the data (0.808 at 75%, 0.806 at 100%), within seed noise. The F1 gap between the models narrows only from 0.089 at 10% to 0.078 at 100%. Within the range we can test, more data of the same kind helps the CNN somewhat but does not close the gap, and the curve gives no evidence of continuing gains; the gap is therefore not explained by training-set size alone. Whether tens of thousands of TCEs, as used by prior deep vetting systems, would change this is outside what these data can show.

Table 8: Test F1 and ROC-AUC vs. training-set size, mean ± std over five seeds.

Train fraction	n_{train}	RF_main F1	CNN_1D F1	RF_main AUC	CNN_1D AUC
10%	314	0.842 ± 0.004	0.753 ± 0.010	0.910 ± 0.004	0.823 ± 0.017
25%	787	0.860 ± 0.006	0.767 ± 0.022	0.928 ± 0.002	0.851 ± 0.016
50%	1,574	0.874 ± 0.004	0.795 ± 0.010	0.936 ± 0.001	0.870 ± 0.006
75%	2,361	0.880 ± 0.006	0.808 ± 0.011	0.941 ± 0.001	0.878 ± 0.004
100%	3,148	0.884 ± 0.004	0.806 ± 0.018	0.942 ± 0.001	0.878 ± 0.012

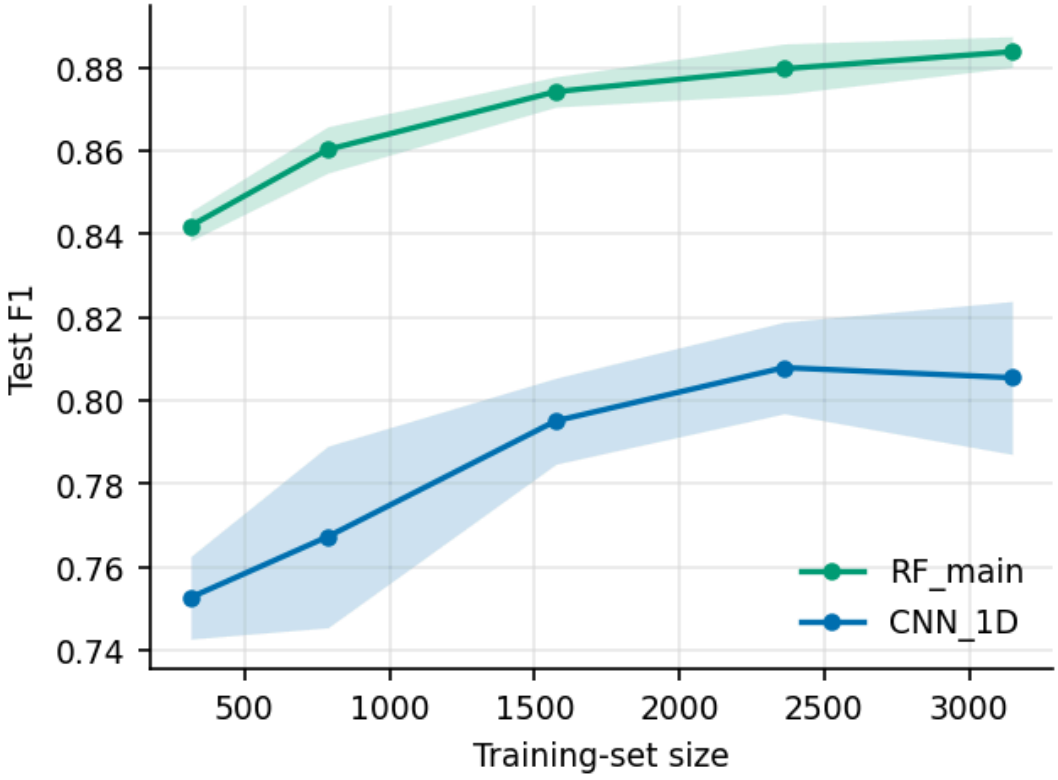


Figure 9: Test F1 vs. training-set size (mean ± std over five seeds).

5.9 Dual-view CNN and ensemble

Table 9 compares CNN_1D with the dual-view architecture (Section 4.5), its 5-model ensemble, and the cosine-schedule control. The dual-view model does not improve F1 or accuracy: F1 is 0.799 ± 0.005 against 0.806 ± 0.018 for CNN_1D and 0.798 ± 0.017 for the cosine-schedule control, and accuracy is 0.810 for both dual-view and CNN_1D. What it does change is run-to-run variance, which is three to four times smaller (F1 std 0.005 vs. 0.018), and slightly better ROC-AUC (0.884 ± 0.003 vs. 0.878 ± 0.012) and Brier score (0.134 vs. 0.138), with about half the parameters (2,049 vs. 3,953). The 5-model ensemble adds nothing in F1 (0.796 ± 0.004). None of these variants approaches RF_main (F1 0.884). This is the expected outcome of a crop that adds no resolution: the local branch re-reads the same few in-transit bins (Section 3.3).

Table 9: CNN_1D vs. dual-view architecture, ensemble and schedule control, test set, mean ± std over five seeds (single dual-view row: 25 runs). RF_main shown for reference.

Model	Params	Accuracy	F1	ROC-AUC	Brier
CNN_1D	3,953	0.810 ± 0.023	0.806 ± 0.018	0.878 ± 0.012	0.138 ± 0.010
CNN_1D (cosine LR)	3,953	0.798 ± 0.027	0.798 ± 0.017	0.871 ± 0.009	0.146 ± 0.010
CNN_dualview	2,049	0.810 ± 0.005	0.799 ± 0.005	0.884 ± 0.003	0.134 ± 0.002
CNN_dualview (5-ens.)	–	0.808 ± 0.003	0.796 ± 0.004	0.885 ± 0.001	0.134 ± 0.001
RF_main	–	0.884 ± 0.003	0.884 ± 0.004	0.942 ± 0.001	0.091 ± 0.001

6 Discussion

On this dataset a Random Forest outperforms a small CNN trained on folded flux, and the gap survives star-grouped splitting, cross-validation, five seeds, and re-weighting to the natural class ratio. The feature-set ablation locates it (Section 5.4). A Random Forest on the four catalog statistics alone (F1 0.880) nearly matches RF_main (0.884), a Random Forest on the nine flux features alone (0.765) falls below the CNN (0.806), and a CNN given

the catalog statistics reaches 0.889, matching RF_main. So the CNN is not worse at learning from this data; it loses because the z-scored 200-bin array does not contain the catalog's period, duration, depth and model SNR, which the Kepler pipeline computes from full-resolution data and which are RF_main's most informative inputs (Figure 5). A nonlinear model over those statistics also matters: logistic regression on the same four statistics reaches only 0.713.

The other diagnostics are consistent with this. Training-set size alone does not account for the gap: the CNN's learning curve flattens and the gap narrows only from 0.089 to 0.078 over a tenfold increase in data. The training recipe does not account for it either, since noise augmentation and a cosine learning-rate schedule give no F1 gain. A cropped local branch gives none because the folded array places a median of only 2.3 bins in transit (Section 3.3), leaving nothing finer to crop; the flux itself carries limited information about transit shape at this resolution, and the CNN's excess errors concentrate on likely eclipsing binaries, where transit-shape and depth-consistency cues would need finer sampling.

The calibration and natural-prevalence results matter for any deployed use. At the natural class ratio even the best fair model, RF_main, produces predicted planet candidates of which about half are false alarms (precision 0.512), so balanced-test F1 overstates operational performance for all models.

The RF_with_robovetter row should not be read as evidence that a Random Forest solves the task almost perfectly. Its near-perfect scores (F1 = 0.991) are explained by the Robovetter confidence score being an input feature, a product of the same disposition logic that produced the labels. It serves as an upper bound and a check that the pipeline is not broken.

7 Limitations and Future Work

First, the lightcurve preprocessing succeeded for only 55.9% of the targeted 8,074 TCEs, primarily due to per-star MAST retrieval failures rather than any systematic exclusion by PC/FP status; the dropout was mild (PC survival 57.7% vs. FP survival 54.2%), leaving a working set of 4,515 TCEs close to balanced (51.6% PC / 48.4% FP). Second, grouped splitting makes partition sizes approximate (661 test TCEs), and the test set is small enough that differences of about one percentage point in F1 are within the seed-to-seed spread of the CNN. Third, the error-mode heuristic compares diagnostics on non-normalized scales, which left two of four intended categories empty. Fourth, we did not tune hyperparameters for either model family, and the hybrid CNN uses a single fixed way of combining the two inputs (concatenation before the head); other fusion designs may behave differently. Fifth, the natural-prevalence metrics assume that recall and false-positive rate measured on the balanced set carry over to the natural population, and the 4,515 successfully downloaded TCEs may differ from the full catalog in ways we did not measure.

Future work should test whether richer flux input adds more once the catalog statistics are available: a true local view re-derived from raw lightcurves at finer time resolution around the transit, together with separate odd and even transit views, would probe this directly. It requires re-downloading the lightcurves and is not attempted here.

8 Conclusion

We diagnosed why a compact 1D CNN underperforms a Random Forest for Kepler DR25 TCE vetting at the scale of 4,515 TCEs. Under star-grouped splits and five seeds, a Random Forest on catalog statistics plus nine engineered flux-shape features (F1 = 0.884 ± 0.004) clearly outperforms the CNN (0.806 ± 0.018) and logistic regression on catalog statistics (0.713); the gap holds under grouped cross-validation (0.901 vs. 0.821) and for every seed pair under McNemar's test ($p < 5 \times 10^{-5}$), and re-weighting to the natural class ratio lowers F1 to 0.649 and 0.533.

Feature-set ablations locate the gap in the inputs rather than in the learner. A Random Forest on the four catalog statistics alone reaches 0.880 ± 0.003, one on the engineered flux features alone reaches 0.765 ± 0.002, below the CNN, and a CNN given the catalog statistics reaches 0.889 ± 0.004, matching the Random Forest. Training-set

size, noise augmentation, a learning-rate schedule and a cropped dual-view branch do not close the gap, and the folded 200-bin input places a median of only 2.3 bins in transit. At the natural class ratio the hybrid CNN's F1 (0.611) remains below the Random Forest's (0.649). A finer-resolution local view with separate odd and even transit views is the main experiment left to test whether the flux can contribute more.

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